

Software Engineering



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Lecture 1

Introduction to Software Engineering

Programming and software engineering



- Software is the programs executed by computers. It may be system software such as operating systems or application software such as games, word-processing, database systems and electronic commerce systems
- Software types(according to Lehman)
 - Static/Specified(S-type): well-defined specification and obvious solution. Example: Calculator program
 - Procedural (P-Type): The controlling procedures are well-defined, but the solution is not so obvious. Example: chess playing
 - Evolving(E-Type): Specification and solution are strongly linked to environment. Example: E-Commerce system

-we have problem & solution in our hands

- we have problem & but solution is not ready now our hands or may be differ from player to another
- Both problem & solution differ around the clock , algorithm and functions update Amazon for example

Software included in P type and e type

What is Software Engineering?

- ❑ **Software Engineering** is the process of designing, developing, testing, and maintaining software. It is a systematic and disciplined approach to software development that aims to create **high-quality**, **reliable**, and **maintainable** software.
- ❑ Software engineering includes a variety of techniques, tools, and methodologies, including requirements analysis, design, testing, and maintenance.

Software engineering & Software engineering projects

- Software engineering is the developing of software product for specific purposes using scientific principles and methods within the constraints of allowed cost, time and other resources.
- Software engineering project is activities and tasks done by a team to produce a software product using the principles and methods of software engineering
- In any engineering project: There should be a plan: What to do? When and how to do it?

Software engineering basic tasks

- What are the basic tasks done in a software project?
 - **Feasibility study:** Is it possible economic project?
 - **Requirement gathering:** Who are the customers? What they want? What they need?
 - **High-level design:** What are the platforms? On premise or on cloud? What are the modules of the system and how they interact without entering in the details of each module.
 - **Low-level design:** assign modules to team members. Design of each module. Module design refinement until it becomes clear how it is implemented in code.
 - **Development:** implementing the design into code by programming the design
 - **Testing:** test your own code, test different parts or code is working with other parts, test if everything is working after integration of all/new modules
 - **Deploying:** Setup hardware, install software, training
- **Maintenance:** While the system is running, collect and solve problems and errors

Feasibility - الامكانية

Platform ex. Site or application ,.....—
deploying for running and publishing for
customer

Engineering software development process

- Why?
 - Because software systems is becoming large and complex.
 - To get scalable system
 - To get cost effective systems in the allowed time
 - To deal with dynamic nature of requirements change and system update
 - To obtain quality in software product
- How?
 - Engineering principles and methods/techniques for doing each of the basic tasks
 - Process model: An overall plan or strategy for starting, working on, finishing all the required basic tasks.

-Every task needs a process and every appeared problem needs update the project through team work

-scalable ..site needs update to enable 1000 user from benefit instead of 100 user for example

Document management system

- While working as a team on the preceding tasks, we need and produce many documents. A document management system is needed to manage these documents. What are the features needed by a document management system for software engineering process?
 - People can share documents so that they can all view and edit them.
 - Only one person can edit a document at a given time.
 - You can fetch the most recent version of a document.
 - You can fetch a previous version of a document by specifying either a date or version number.
 - You can search documents for tags, keywords, and anything else in the documents.
 - You can compare two versions of a document to see what changed, who changed it, and when the change occurred. (Ideally, you should also see notes indicating why a change was made, although, that's a less common feature.)
 - You should have the ability to access documents over the Internet or on mobile devices.

Feasibility study

- Identifying the problem, requirements, defining the objectives/goals, scope and expected outcomes.
- Search alternatives
- Assessing Technical Feasibility
- Assessing Economic Feasibility
 - Assessing Operational Feasibility
 - Assessing Legal Feasibility
 - **Schedule Feasibility**

Operational: manual operation and ideas
can be made as automatically

Systems classification

- With respect to feasibility study, software systems can be categorized:
- Replacement or new?
 - A system that replaces an existing computer-based system
 - A brand-new system that replaces or enhances work that is not currently computer-assisted
- General/Special purposes
 - A general-purpose system, such as a word processor or a game. This is written and then sold in the marketplace
 - A tailor-made one-off system for a specific application.
- Develop or Buy
 - Writing the software or buying it off the shelf?

Technical feasibility

- Is the proposal technically feasible? That is, will the technology work?
- Examples:
 - A system to predict lottery results cannot work. BUT CAN EXPECT
 - A system to understand natural-language dialog is borderline محدود
 - A system that manage a store is possible

Cost-benefit analysis



- Comparing:
 - The cost of providing the system- Cost
 - The money saved or created by using the system – the benefit
- Is benefit passes the cost? For more than one way to do the work: which way you choose? The greater benefit/cost ratio
- Period: cost-benefit for which period?
 - 5 years to overcome and distribute the starting cost
- Which cost?
 - cost to buy equipment, principally the hardware
 - Cost to develop the software
 - Cost of training
 - Cost of lost work during switchover
 - cost to maintain the system
 - Cost to repair the equipment in the event of failure
 - Cost of lost work in the event of failure
 - Cost to upgrade, in the event of changed requirements.

Example Feasibility study of the software for an ATM

- Required:
 - 200 ATM systems for a bank each is complete with its hardware and software
- Technical feasibility
 - It is possible to build ATMs using available hardware components. The software can be specially designed or bought
- Cost-benefit study
 - Hardware cost
 - System parts:
 - Hardware: For each ATM: the processor, the card reader, the display, the screen, the keypad, the printer, a cash dispenser and a modem
 - AT the bank server and other resources
 - Startup cost تكلفه انشاء
 - ATM machine hardware: 10000
 - Server and other recourses the bank to respond to ATM requests=2000/ATM-machine
 - Installation in secure way=8000/ATM-machine
 - Running cost: تكلفه تشغيل واستهلاك
 - telephone-bill, changing printer paper, r, stocking the ATM with cash.=1000/ATM-machine/year

Example Feasibility study of the software for an ATM-Continued

- Cost-benefit study
 - Software cost: The ATM software must be robust and reliable because it is used by members of the public
 - System parts:
 - ATM machine software system
 - Bank server system
 - Startup-cost
 - An estimate: two person years for the software=450/Atm-machine
 - Running cost
 - Update and enhancements= 50/Atm-machine/year

Example Feasibility study of the software for an ATM-Continued

- Cost-benefit study

- Benefit

- ATMs are convenient, available 24/7 hours in stations, stores and public buildings. So they attract customers to bank=1000/ATM-machine/year
 - Replacing employees and bank space=40000/ATM-machine/year

- Cost-benefit summary

- For five years period

- Cost/ATM-machine/year= $21000(\text{Hardware}) + 500(\text{software}) = 21500$
 - Benefit: 40000 ATM-machine/year

- Result:

Should the ATMs be developed? YES

Running example: Library management

System: Description:

It is needed to automate a currently manual managed • library. The library receive victors for searching for books, read books inside the library. A visitor can ask to be registered as library-member or client. A library member can borrow books to read them at home

Today task (Report):

Do a feasibility study for the suggested system •